Hao Fang

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Google Scholar | Personal Website

Date of birth: 6 May 1999 | Citizenship: China | Highest degree: PhD expected May 2027

Research Interests

Computational Imaging, Intelligent Sensing, Digital Twins, Computer Vision

Education

- **The University of Edinburgh** *Oct 2023 – Present*
  PhD in Engineering (expected May 2027), School of Engineering Edinburgh, UK
  - Research on clinically ready, structure-aware imaging of pulmonary function dynamics enabled by digital respiration-driven learning.
- **The University of Edinburgh** *Sep 2021 – Dec 2022*
  MSc in Signal Processing and Communications Edinburgh, UK
- **Southwest Jiaotong University** *Sep 2017 – Jun 2021*
  BEng in Electronic and Electrical Engineering, SWJTU–Leeds Joint School Chengdu, China
  - Dual-degree programme with the University of Leeds; weighted average: 90.36/100; top 3% of the cohort.
- **University of Leeds** *Sep 2017 – Jun 2021*
  BEng in Electronic and Electrical Engineering Leeds, UK
  - First-class honours degree; completed an exchange period on the Leeds campus.

Professional and Research Experience

- **Jiangsu Industrial Technology Research Institute [🌐]** *Nov 2022 – Sep 2023*
  Software and Algorithm Engineer, JITRI Brain–Machine Fusion Intelligence Institute Suzhou, China
  - Developed and implemented software systems and algorithms for brain–computer interfaces (BCIs).
  - Supervised master's students jointly trained by Xi'an Jiaotong–Liverpool University and the institute, contributing to research design, algorithm development, and technical guidance for their dissertations.
- **The University of Edinburgh [🌐]** *Mar 2024 – Present*
  Research Assistant Edinburgh, UK
  - Contributed to research and technical development; served as a Tutor and Demonstrator for undergraduate and master's courses, providing tutorials, laboratory instruction, and technical guidance to students.
- **Academic Service** *2023 – Present*
  Journal Reviewer
  - Reviewed for IEEE journals including *IEEE Transactions on Image Processing*, *IEEE Transactions on Industrial Informatics*, *IEEE Transactions on Instrumentation and Measurement*, and *IEEE Transactions on Computational Imaging*.

Projects

- **Precision Four-Dimensional Pulmonary Function Imaging (Sim2Clinic Framework)** *2023 – Present*
  Research project: medical imaging algorithms, deep learning, and digital twins
  - Built a Sim2Clinic data-generation pipeline for 4D pulmonary electrical impedance tomography (EIT): segmented lungs, thoracic cavities, and other tissue structures from a large set of clinical CT scans; created patient-specific finite-element models and electrode layouts in COMSOL; and generated a large-scale 4D EIT dataset using respiratory physiology and dynamic simulations.
  - Developed a modular training strategy, multimodal single-frame reconstruction models, and temporal autoregressive deep reconstruction models to reconstruct 3D/4D pulmonary function images from low-dimensional voltage sequences.
  - Modelled realistic measurement perturbations and developed patient-adaptive strategies; assessed robustness and generalisability using multicentre clinical data.
- **SELECT: Next-Generation Soft Robotics** *2024 – Present*
  Research project: robotic perception algorithms
  - Funded by a European Research Council (ERC) Starting Grant (No. 101165927).
  - Investigated EIT reconstruction algorithms to improve tactile and proprioceptive sensing in soft robotic systems.
- **FLEX-AI: Scalable Electronic Skin for Autonomous Soft-Robot Intelligence** *2026 – Present*
  Research project: algorithms and perception
  - Funded by an ERC Proof of Concept Grant (No. 101292593).
  - Developed perception algorithms for soft robots.

- **Major Non-Communicable Chronic Disease Imaging Project** *2024 – Present*
  Research project: algorithms and biomedical imaging; Peking Union Medical College Hospital
  - Supported by a National Science and Technology Major Project (No. 2024ZD0522700).
  - Developed medical imaging algorithms for continuous respiratory-function monitoring, focusing on pulmonary function assessment and bedside monitoring.
- **Real-Time Bedside Medical Imaging System for Intensive Care** *2024 – Present*
  Industry–academia collaborative project: image reconstruction algorithms
  - Joint research project between Prunus Medical and the University of Edinburgh.
  - Investigated three-dimensional pulmonary EIT algorithms for real-time bedside imaging in intensive care units (ICUs).
- **BrainTunes: Closed-Loop BCI Music Interaction Mini-Program** *2022 – 2023*
  Industrial project: software and algorithm development
  - Designed and implemented a lightweight WeChat mini-program for a closed-loop BCI device, integrating physiological-signal acquisition and visualisation with music-based closed-loop feedback.

Publications and Patents

J = journal article, C = conference paper, P = patent

Published and accepted papers

[J.1] **H Fang**, et al. (2026). **SBGC-θ: Self-Balancing Gradient Coordination for EIT Monitoring Systems**, *IEEE Transactions on Industrial Informatics*. Accepted for publication; DOI: 10.1109/TII.2026.3743018; first author; JCR Q1; IF 9.8.

[J.2] **H Fang**, et al. (2026). **QuantEIT: Ultra-Lightweight Quantum-Inspired Inference for Electrical Impedance Tomography**, *IEEE Transactions on Industrial Informatics*. Accepted for publication; DOI: 10.1109/TII.2026.3731100; first author; JCR Q1; IF 9.8.

[J.3] **H Fang**, et al. (2026). **Deep Dynamic Image Prior for Three-dimensional Time-sequence Pulmonary Electrical Impedance Tomography**, *Engineering Applications of Artificial Intelligence*. DOI: 10.1016/j.engappai.2025.113521. First author; JCR Q1; IF 9.0.

[J.4] **H Fang**, et al. (2025). **Multifrequency Electrical Impedance Tomography Reconstruction with Multibranch Attention Image Prior**, *IEEE Internet of Things Journal*. DOI: 10.1109/JIOT.2025.3624228. First author; JCR Q1; IF 8.7.

[J.5] L Li, **H Fang**, et al. (2026). **Self-Supervised Cross-Source Point Cloud Registration Framework with Geometric Tree Transformer for HMD 6D Pose Estimation**, *IEEE Transactions on Industrial Electronics*. DOI: 10.1109/TIE.2026.3670314. Second author; JCR Q1; IF 7.4.

[J.6] Z Liu, Z Chen, **H Fang**, et al. (2025). **Regularized Shallow Image Prior for Electrical Impedance Tomography**. *IEEE Transactions on Instrumentation and Measurement*. DOI: 10.1109/TIM.2025.3545548. JCR Q1; IF 7.0.

[C.1] **H Fang**, et al. (2026). **Dual-branch Attention Regularization for Clinical 3D Pulmonary Electrical Impedance Tomography**. *IEEE I2MTC 2026*. Oral presentation.

[C.2] **H Fang**, et al. (2024). **Multi-Modal EIT Imaging Using Lensfree and Flexible Impedance Sensor**. *2024 IEEE SENSORS*. DOI: 10.1109/SENSORS60989.2024.10784633.

[C.3] **H Fang**, et al. (2024). **Frequency-difference Cell Imaging with Flexible Micro-EIT Sensor**. *EIT 2024*.

[C.4] **H Fang**, et al. (2023). **V-SWIR-IF: Visible and Short-Wave Infrared Image Fusion**. *ISCEIC 2023*. DOI: 10.1109/ISCEIC59030.2023.10271197.

[C.5] S Teng, **H Fang**, et al. (2026). **Deformation-Insensitive EIT Electronic Skins via Honeycomb Lattices**. *IEEE I2MTC 2026*. Oral presentation.

[C.6] Y Li, **H Fang**, et al. (2024). **Minimal Electrode EEG for BCI Emotion Detection**. *NNICE 2024*. DOI: 10.1109/NNICE61279.2024.10499167.

[C.7] G Xu, G Su, **H Fang**, et al. (2023). **Enabling Transcranial Electrical Stimulation via STI01: Experimental Simulations and Hardware Circuit Implementation**. *EEI 2023*. DOI: 10.1109/EEI59236.2023.10212634.

Under review, in revision, and in progress

[J.7] Z Liu*, **H Fang***, et al. (2026). **Physiologically realistic *3D+t* lung ventilation imaging**. Co-first author; planned submission to *Nature* in Dec 2026.

[J.8] **H Fang**, et al. (2026). **MESA: Measurement-Driven Estimation with Spatial Covariance and Adaptive Contrast Readout for Inverse Imaging**. *IEEE Transactions on Image Processing*. First author; under review; JCR Q1; IF 15.3.

[J.9] **H Fang**, et al. (2026). **Robust three-dimensional pulmonary electrical impedance tomography via training-free dual-frame network prior**. *Engineering Applications of Artificial Intelligence*. First author; under review; JCR Q1; IF 9.0.

[J.10] **H Fang**, et al. (2026). **AnatoEIT: Reusable CT–EIT Fusion for Volumetric Pulmonary Imaging with Quantified Prior-Mismatch Limits**. *Information Fusion*. First author; under review; JCR Q1; IF 17.4.

[J.11] **H Fang**, et al. (2026). **Accelerating Iterative EIT Reconstruction for Real-Time Tactile Sensing**. *IEEE/ASME Transactions on Mechatronics*. First author; under review; JCR Q1; IF 6.3.

[J.12] **H Fang**, et al. (2026). **Fusing locally adaptive soft-clique construction with support–coverage consensus for robust 3D Registration**. *Information Fusion*. First author; under review; JCR Q1; IF 17.4.

[J.13] **H Fang**, et al. (2026). **BRF: Bayesian-Guided Regional Fitting for Inverse Imaging**. *Pattern Recognition*. First author; under review; JCR Q1; IF 9.1.

[J.14] L Li, ..., **H Fang*** (2026). **DAVD: Density-Aware Versatile Descriptors and Progressive Correspondence Pruning for Robust Point Cloud Registration**. *IEEE Transactions on Pattern Analysis and Machine Intelligence*. Corresponding author; under review; JCR Q1; IF 18.6.

[J.15] Z Liu, ..., **H Fang**, et al. (2026). **Monolithic perceptive soft machines via volumetric body-field coupling**. *Nature Sensors*. Under review.

[J.16] S Teng, **H Fang**, et al. (2026). **Flexible Electrical Impedance Tomography for Tactile Sensing with Deformation Prior**. *IEEE Transactions on Industrial Electronics*. Second author; major revision; JCR Q1; IF 7.4.

[J.17] S Teng, L Li, **H Fang**, et al. (2026). **Stretch-Tolerant Tactile Sensing through Mechanically Architected EIT E-Skins**. *Soft Science*. Major revision; JCR Q1; IF 15.9.

[J.18] Z Feng, C Wang, **H Fang**, et al. (2026). **Agentic LLM–RAG Orchestration for Longitudinal Lung-RADS Classification from Unstructured Chinese CT Report Findings**. *Engineering Applications of Artificial Intelligence*. Under review; JCR Q1; IF 9.0.

[C.8] **H Fang**, et al. (2026). **LASC: Locally Adaptive Soft-Clique Construction and Support–Coverage Consensus for Robust 3D Registration**. *ICRA 2027*. First author; under review; CCF-B.

[C.9] L Li, S Teng, **H Fang**, et al. (2026). **Temporal–Static Fusion for Dense Soft-Body Proprioception with Stretchable EIT Skin**. *ICRA 2027*. Under review; CCF-B.

Granted patents

[P.1] **Multichannel Electroencephalogram Signal Acquisition Device**. Chinese utility-model patent CN219147613U (2023).

[P.2] **Integrated Active Dry Electrode and EEG Acquisition Device**. Chinese utility-model patent CN219270940U (2023).

Honours and Awards

- **Peter Grant Best Paper Prize (Best Conference Paper)** *2026*
  Institute for Imaging, Data and Communications (IDCOM), University of Edinburgh
- **I2MTC 2026 Student Travel Grant (USD $1,200)** *2026*
  IEEE Instrumentation and Measurement Society
- **2025 Outstanding Reviewer, IEEE Transactions on Instrumentation and Measurement** *2025*
  IEEE Instrumentation and Measurement Society
- **Outstanding Student** *2020*
  Southwest Jiaotong University
- **First-Class Comprehensive Scholarship (Top 3%)** *2020*
  Southwest Jiaotong University